

# Effect of Antioxidant Grafting on Conductivity and Carrier Migration Characteristics of Polypropylene Composite Cable Insulation

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**Abstract**—To investigate the effect of grafting modification on the carrier migration characteristics of polypropylene (PP) elastomer composite insulation, a composite insulation of antioxidant grafted modified PP blend thermoplastic elastomer is prepared. The evolution laws of direct current (dc) conductivity and dc integral charge [DCIC- $Q(t)$ ] of the composite insulation are studied, and the influence mechanism of grafting modification on the distribution characteristics of charge traps is analyzed. Research has found that compared to unmodified composite insulation, graft-modified composite insulation can reduce leakage current by 33.4% under high temperature and high electric field, increase the threshold field for charge injection by 30%–50%, significantly reduce the accumulation of space charge, and increase the trap energy level, exhibiting stronger trapping of carriers. Through experiments and simulations, it was found that grafting not only introduced localized deep trap states in the energy levels of PP, increasing the trap density, but also enhanced the intermolecular interaction energy, thereby restricting carrier migration under high temperatures and high electric fields, ultimately improving the insulating performance of the material.

**Index Terms**—Carrier migration, conductivity, dc integral charge [DCIC- $Q(t)$ ], grafting modification, polypropylene (PP) elastomer composite insulation.

## I. INTRODUCTION

HIGH-VOLTAGE cables are the main means of power transmission over long distances. High-voltage cables play an irreplaceable and important role in the reliability, efficiency, economy, and environmental protection of power systems. Compared with traditional crosslinked polyethylene (XLPE) cables, polypropylene (PP), a thermoplastic material, has gained considerable attention due to its outstanding insulating and heat-resistant performance, as well as the advantage that it does not require crosslinking or degassing. [1], [2], [3], [4].

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However, PP has poor toughness, and in cold environments, cable insulation made of PP may break due to mechanical impact or bending, resulting in damage to the cable and interruption of power transmission. Therefore, there is a need to modify the toughness of PP. Zhou et al. [5] blended PP with the polyolefin elastomer (POE) to enhance the mechanical properties of PP and found that the introduction of POE improved the flexibility of PP but made the direct current (dc) breakdown strength of the material slightly reduced and the accumulation of heteropolar space charge enhanced. Hosier et al. [6] investigated a series of propylene copolymers with respect to their mechanical and electrical properties and found that with the increase in the number of elastic phases, the dc breakdown properties decreased with an increase in the elastic phase. It can be found that when adding elastomers or introducing elastic phases into PP, the insulation performance of the material will decrease, making it impossible to achieve comprehensive control of the mechanical and electrical properties of the material. Therefore, it is necessary to modify it [7], [8], [9].

Polymer graft modification is a method of introducing polar or functional side groups onto the polymer macromolecular chain. It avoids problems such as aggregation of nanoparticles and migration and diffusion of blended molecules and has great potential for application in high-voltage cable insulation. Graft modification technology has been applied in recent years to enhance the insulation performance of cable materials. Studies have shown that PP materials grafted with monomers like styrene, maleic anhydride, and silane demonstrate outstanding insulation performance [10], [11], [12], [13], [14], [15].

However, current research on grafted modified PP materials for cable insulation has not considered the addition of elastomers. The mechanism of the effect of grafting on the migration of carriers in PP composite insulations under high temperature and high electric field is not yet clear. Further investigation into the effects of grafting modification on carrier migration in PP elastomer composite insulation materials under different electric fields is of significant engineering importance.

Antioxidants are commonly used additives in cable manufacturing; therefore, an antioxidant was selected as the grafting monomer in this article. The evolution laws of dc conductivity, space charge, and trap characteristics of composite insulation

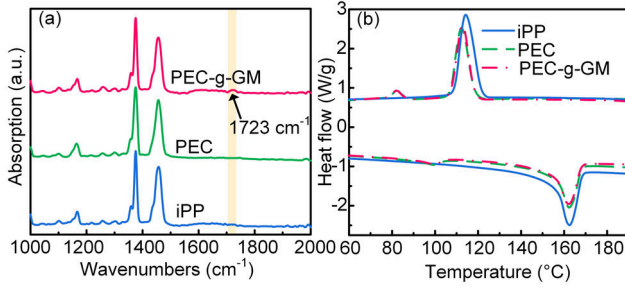


Fig. 1. (a) FTIR spectra of samples. (b) DSC curves of samples.

were studied, and the energy-level structure of molecules was calculated. The influence mechanism of grafted antioxidant on the migration characteristics of electron and hole carriers was analyzed.

## II. EXPERIMENTAL SECTION

### A. Sample Preparation

The selected PP was T30S isotropic PP (iPP) produced by Sinopec Maoming Company. The elastomer was POE 8480 from Sinopec Maoming Company. The hindered phenolic antioxidant 2-tert-butyl-6-(3-tert-butyl-2-hydroxy-5-methylbenzyl)-4-methylphenyl acrylate (GM), the initiator benzoyl peroxide (BPO), and acetone were purchased from Shanghai Macklin Biochemical Technology Company Ltd.

A certain amount of PP granules was placed in xylene solvent and fully dissolved at 120 °C; then, 1 wt% of GM was added [14], and ultrasonic shaking was carried out at 120 °C for 10 min to make GM uniformly dispersed. After the temperature was lowered to 100 °C, 0.33 wt% BPO was added according to the reaction ratio, and the reaction was carried out under the protection of nitrogen atmosphere for 2 h. After that, the mixture was filtered and vacuum dried at 80 °C for 24 h to obtain the PP grafting powder, and then, the powder was washed with acetone to further purify the grafting product. Finally, the grafted product and POE were blended together in xylene solvent to obtain the graft-modified PP elastomer composite named PEC grafted GM insulation (PEC-g-GM), in which the mass ratio of PP to POE was 7:3. The composite material without grafting was named polypropylene elastomer composite (PEC). The grafting percentage (GP) was calculated by the following equation:

$$GP = (w_1 - w_0)/w_1 \times 100\% \quad (1)$$

where  $w_0$  is the mass of PP before grafting and  $w_1$  is the mass of the grafted sample after cleaning with acetone. The reaction grafting rate was calculated to be 0.71 wt%.

This article used Fourier-transform infrared (FTIR) spectroscopy to verify the success of the grafting process, as shown in Fig. 1(a). It can be seen that comparing PP and PEC, PEC-g-GM shows a characteristic peak at 1723  $\text{cm}^{-1}$ . This peak corresponds to the characteristic absorption peak of carbonyl group. Because of the presence of carbonyl functional group in GM, it can be judged that the grafting reaction occurred successfully.

The melting and crystallization curves of each sample are shown in Fig. 1(b). It can be seen that compared with iPP, PEC and PEC-g-GM showed crystallization and melting peaks

TABLE I  
MELTING AND CRYSTALLIZATION PARAMETERS OF SAMPLES

| Sample   | $T_m$ (°C) | $T_c$ (°C) | $X_c$ (%) |
|----------|------------|------------|-----------|
| iPP      | 162.49     | 115.73     | 47.61     |
| PEC      | 162.35     | 114.19     | 43.74     |
| PEC-g-GM | 162.08     | 115.35     | 42.40     |

around 80 °C and 100 °C, respectively, which were related to POE. The PP melting temperatures  $T_m$  of the three samples around 162 °C were very close to each other, indicating that grafting did not significantly change the melting characteristics. It can be seen that the crystallization temperature  $T_c$  of PP in PEC is lower than that of PP, which is due to the fact that the mixed POE disrupts the molecular chain arrangement of PP, making PP more difficult to crystallize.  $T_c$  of PEC-g-GM was slightly higher than that of PEC, which was related to the heterogeneous nucleation of the grafted GM. In addition, the crystallinity of each sample was calculated according to the following equation, as shown in Table I:

$$X_c = \frac{\Delta H_m}{\omega H_m^0} \times 100\% \quad (2)$$

where  $X_c$  is the degree of crystallinity,  $\Delta H_m$  is the enthalpy of melting,  $H_m^0$  is the standard enthalpy of melting, and the standard enthalpy of melting of iPP is 209 J/g.  $\omega$  is the mass fraction of PP in the mixed insulation sample. Compared to PEC, the crystallinity of PEC-g-GM is reduced, which is mainly due to the amorphous structure of the GM molecular chain, thus resulting in a reduced percentage of the PP crystalline region.

The crystalline morphology of the samples was observed using polarized optical microscopy (POM), as shown in Fig. 2. As seen in the images, the spherulites in PEC appear more diffuse compared to iPP, due to the increased amorphous regions in the sample caused by the addition of POE, although the spherulite size remains close to that of iPP. In contrast, the spherulite size in PEC-g-GM is significantly smaller, as the heterogeneous nucleation effect of the GM monomer generates more nuclei, resulting in the formation of a larger number of spherulites [16]. Smaller spherical crystals lead to more interfaces between the spherical crystals, thus increasing the number of traps [17], which is corroborated by the later experiments on trap energy levels.

The stress-strain behavior of the PP materials was analyzed and compared, as shown in Fig. 3. Pure PP exhibits a very low elongation at break, indicating its poor toughness. The PEC composite shows a significantly higher elongation at break than PP, suggesting that the incorporation of POE improves the toughness of the material. After graft modification, the elongation at break of PEC-g-GM slightly decreases, which can be attributed to enhanced interfacial interactions introduced by the grafted groups, restricting the mobility of the elastomer phase. Nevertheless, due to the low grafting content, the elongation at break of PEC-g-GM remains much higher than that of pure PP.

### B. Experimental Setup

The leakage current of the sample was measured by means of a three-electrode system at different electric fields

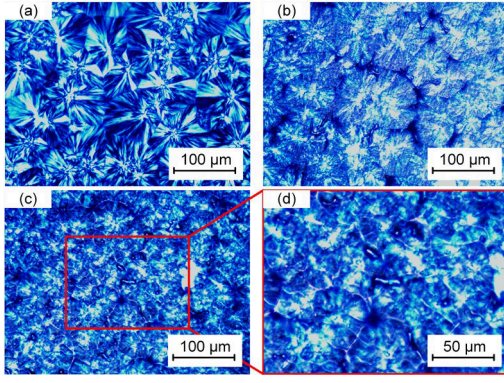


Fig. 2. POM results of samples. (a) iPP. (b) PEC. (c) PEC-g-GM. (d) Enlarged view.

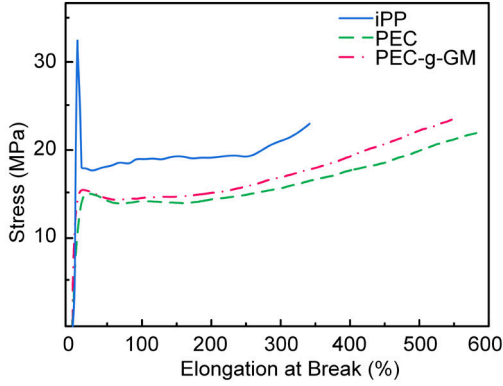


Fig. 3. Stress-strain curves of different materials.

(1–70 kV/mm) and at different temperatures (30 °C–90 °C) according to IEC 60 093. The samples were thin sheet samples of 150 μm in thickness. The steady state current after 30 min of polarization was selected as the measured value.

The total charge within the material over time was measured using the dc integrated charge [DCIC– $Q(t)$ ]. The measured electric fields were 1, 5, 10, ..., increasing in increments of 5 kV/mm, up to 70 kV/mm. After polarizing the sample under an electric field for 10 min, the sample was removed, and the electrodes were discharged for 5 min. Then, a new sample was placed for measurement at the next electric field. The measurement curves were fit with the three-branch model. The components are shown in the following equation [18]:

$$Q(t) = Q_0 + Q_{\text{abs}} \left[ 1 - \exp\left(-\frac{t}{\tau}\right) \right] + I_{\text{cond}} \cdot t \quad (3)$$

where  $Q_0$  is the initial charge;  $Q_{\text{abs}}$  is the absorbed charge, which contains the insulating dielectric polarization process and the charge injection process;  $I_{\text{cond}}$  is the conduction current, which describes the dc conductivity of the dielectric; and  $\tau$  is the relaxation time constant.

The trap energy-level distributions of each sample were obtained by the thermally stimulated depolarization currents (TSDCs) method [19]. The samples were initially polarized at 70 °C and 30 kV/mm for 10 min, then quickly cooled to –50 °C, and short-circuited for 5 min to remove the influence of stray charges. Subsequently, they were heated from –50 °C

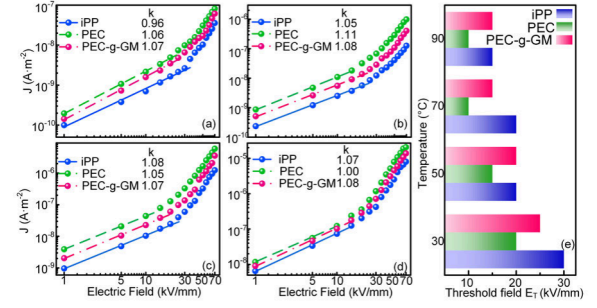


Fig. 4. Fitting results of ohmic conductivity mode. (a) 30 °C. (b) 50 °C. (c) 70 °C. (d) 90 °C. (e) Threshold electric fields  $E_T$ .

to 100 °C at a rate of 3 °C/min while recording the current profiles.

### III. EXPERIMENTAL RESULTS

#### A. Conductivity Characteristics

The variation of the current density  $J$  of the samples with the applied dc electric field strength  $E$  and temperature  $T$  is shown in Fig. 4. The current density increases significantly with increasing temperature or electric field. It is observed that the current density of PEC consistently exceeds that of iPP, suggesting that the incorporation of POE reduces the insulating performance of iPP. In addition, although the current density of PEC-g-GM is also higher than that of iPP, the rise is significantly decreased compared to PEC, with the highest decrease reaching 59.02%. It indicates that the grafted GM inhibits the carrier migration inside iPP.

This article focuses on the changes in the conductivity mechanism of graft-modified composite insulating materials at different temperatures and electric fields, where the conductivity mechanism is influenced by space charge. It primarily includes the ohmic conductivity, hopping conductivity, and Poole–Frenkel (P–F) emission. Additionally, under high temperature, carriers can overcome the contact barrier between the metal electrode and the insulating matrix, leading to injection phenomena. Therefore, Schottky emission is also a key factor influencing insulation conductivity under these conditions.

Under the influence of an external electric field, electrons move directionally within the conduction band, generating an ohmic conduction current. The current density of ohmic conductivity satisfies the following equation [20]:

$$J = \sigma E = nq\mu E \quad (4)$$

where  $J$  is the current density,  $\sigma$  is the conductivity,  $n$  is the number of carriers,  $q$  is the carrier charge,  $\mu$  is the carrier mobility, and  $E$  is the electric field. The current generated by ohmic conductivity is usually small and only dominates at lower temperatures and electric fields. The conformity of the dominant conductivity mechanism to ohmic conductivity can be determined by whether the slope  $k$  of the fit curve of  $\ln J$  versus  $\ln E$  is 1 or not. The fitting results are shown in Fig. 4.

It can be seen that the fit slopes  $k$  for all three samples are very close to 1 at low fields, but as the electric field rises, the slopes gradually become larger than 1, indicating that the conductivity of the dielectric is no longer dominated by ohmic conductivity but by other conductivity mechanisms.

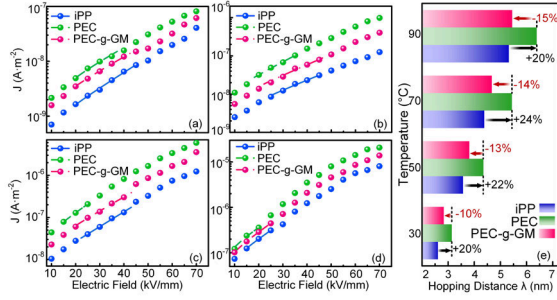


Fig. 5. Fitting results of hopping conductivity model. (a) 30 °C. (b) 50 °C. (c) 70 °C. (d) 90 °C. (e) Hopping distance.

The threshold field  $E_T$  was defined as the highest electric field at which ohmic conductivity behavior is still maintained. As shown,  $E_T$  for PEC-g-GM is consistently higher than that of PEC across all tested temperatures, and is even comparable to that of iPP. This indicates that PEC-g-GM exhibits a broader ohmic conductivity region than PEC, reflecting its improved insulation stability under elevated fields. The increase in the threshold electric field of PEC-g-GM may be related to its trap energy levels and trap density, which is confirmed in Fig. 4.

### B. Hopping Conductivity

In the low-to-medium electric fields, the energy obtained by carriers is low, and carriers migrate between localized states in a hopping manner, forming hopping conductivity inside the insulation. The current density generated by hopping conductivity satisfies the following equation [21]:

$$J = 2nq\lambda v \exp\left(-\frac{E_a}{k_B T}\right) \sinh\left(\frac{\lambda q E}{2k_B T}\right) \quad (5)$$

where  $n$  is the carrier concentration,  $q$  is the charge,  $\lambda$  is the average hopping distance,  $v$  is the thermal vibration frequency,  $E_a$  is the hopping conductivity activation energy,  $k_B$  is Boltzmann's constant, and  $T$  is the absolute temperature. Under certain temperature conditions, (5) can be simplified and described as follows:

$$J = \alpha \sinh(\beta E) \quad (6)$$

where  $\alpha$  and  $\beta$  are constants.

From Fig. 4, it can be seen that below 10 kV/mm is dominated by ohmic conductivity, so the fit starts from 10 kV/mm, and the results are shown in Fig. 5(a)–(d).

The hopping conductivity of PEC is more significant when the electric field is lower than 35 kV/mm under the temperature condition of 30 °C. As the temperature increases to 90 °C, the conductivity mechanism shows an obvious form of hopping conductivity when the externally applied field is lower than 20 kV/mm. This indicates that as the temperature increases, the electric field region dominated by hopping conduction gradually decreases. However, the hopping conductivity mechanism of PEC-g-GM at 90 °C is still dominant at field strengths lower than 25 kV/mm and has the same level as PP at other temperatures.

The hopping distances of the samples were calculated by (5) and shown in Fig. 5(e). The hopping distance in PP insulation is influenced by test temperature, applied electric

field, crystallinity, and morphological structure. In this study, the hopping distance is used as an indicator to evaluate the insulation performance of PP [22], [23]. With increasing temperature and electric field, the hopping distance exhibits a pronounced increase. It is evident that the PEC insulation demonstrates a significantly larger hopping distance compared to the other two materials. This can be attributed to the fact that POE predominantly exists in the amorphous phase, where the molecular chains are more loosely arranged than in the crystalline phase, thereby facilitating charge carrier migration. The hopping distance of PEC-g-GM is shorter than that of PEC and closely approaches that of PP. Deeper and denser traps result in a shorter hopping distance. Therefore, it can be inferred that grafting GM increases the trap energy level in PEC, making it more difficult for charge carriers to overcome the traps under the same conditions. This leads to a reduction in the hopping distance and effectively suppresses carrier transport in PEC-g-GM.

### C. Schottky Emission and P–F Emission

There exists a Schottky emission effect between the electrode and the insulation under high temperature and strong electric fields. This is because the carriers are excited by heat to obtain sufficient energy to overcome the interface potential barrier, resulting in the process of injection. The current density generated by Schottky emission can be expressed as follows [24]:

$$J = AT^2 \exp\left(\frac{-q(\phi_B - \sqrt{qE/4\pi\epsilon_r\epsilon_0})}{k_B T}\right) \quad (7)$$

where  $A$  is the Richardson constant,  $T$  is the absolute temperature,  $q$  is the charge of carriers,  $q\phi_B$  is the Schottky injection barrier,  $\epsilon_r$  is the relative dielectric constant,  $\epsilon_0$  is the vacuum dielectric constant,  $k_B$  is Boltzmann's constant, and  $E$  is the electric field. The fitting results based on Schottky emission are shown in Fig. 6(a)–(d). Although the data points are all better fit with a linear relationship, conformity to the Schottky injection model still needs to be judged by the dielectric constant inverted by the slope of the straight line. Taking (7) logarithmically, the dielectric constant calculated from the slope is shown in Fig. 6(e). The dielectric constants of PP and POE are both approximately 2.4, which is therefore used as the reference value for Schottky emission analysis. At 30 °C and 50 °C, the fit dielectric constants for all samples are close to 2.4, suggesting that Schottky emission is the dominant conduction mechanism in this temperature range. The minimum electric field along the fit line that satisfies the Schottky emission model is defined as the threshold electric field. Notably, the threshold electric field for PEC-g-GM is higher than that for PEC, indicating that charge injection is more difficult in PEC-g-GM. However, at elevated temperatures (70 °C and 90 °C), the fit dielectric constant deviates significantly from the reference value, implying that Schottky emission is no longer the sole dominant mechanism under these conditions. It is considered to be caused by the P–F emission. The P–F emission can be formulated as follows [25]:

$$J = q\mu N_c E \exp\left(\frac{-q(\phi_T - \sqrt{qE/\pi\epsilon_r\epsilon_0})}{k_B T}\right) \quad (8)$$

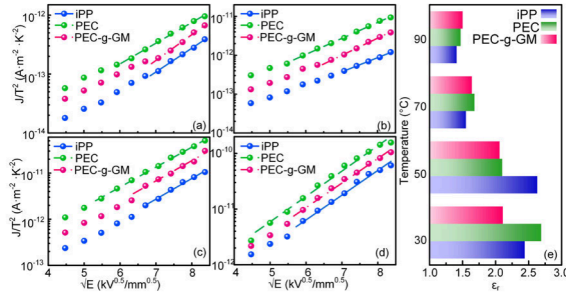


Fig. 6. Fitting results of Schottky emission model. (a) 30 °C. (b) 50 °C. (c) 70 °C. (d) 90 °C. (e) Fitting results  $\epsilon_r$ .

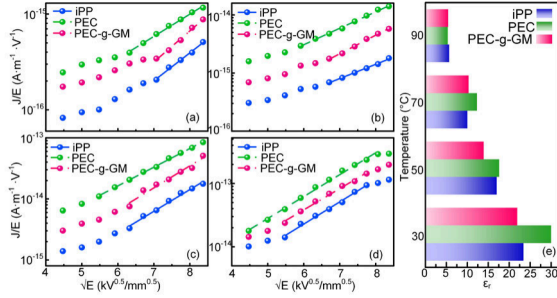


Fig. 7. Fitting results of P-F emission model. (a) 30 °C. (b) 50 °C. (c) 70 °C. (d) 90 °C. (e) Fitting results  $\epsilon_r$ .

where  $\mu$  is the carrier mobility,  $N_c$  is the density of states in the conduction band, and  $q\Phi_T$  is the activation energy for P-F emission. The other parameters have the same meanings as those in the Schottky emission formula. The fitting results are shown in Fig. 7. It is apparent that the fitting results show a significant deviation from the dielectric constant. However, as the temperature increases, the fitting result gradually approaches 2.4, indicating that the influence of P-F emission on conductivity is gradually increasing. By combining with the fitting results for Schottky emission, it can be concluded that the conductivity at high electric fields at 70 °C and 90 °C is primarily affected by the synergistic effect of these two mechanisms.

#### D. Dominant Conductivity Mechanism

Based on the previous fitting results and conclusions, along with the threshold fields for ohmic conductivity and Schottky emission, distribution maps of the dominant conductivity mechanisms of the samples at various temperatures and electric fields were acquired, as shown in Fig. 8(a)–(c). At low temperatures and electric fields, conduction is mainly dominated by ohmic conduction. As the temperature increases, charges undergo hopping transport through traps, and conduction is then dominated by hopping conduction. With the further increase in temperature and electric field, charges overcome the interfacial barrier between the electrode and the sample and are injected into the sample, at which point conduction is dominated by Schottky injection. Focusing on the comparison of the material's electrical performance under high temperatures and strong electric fields, it can be observed that the region dominated by Schottky emission and P-F emission for PEC-g-GM is much smaller than that for PEC, and it is very similar to that of iPP. This indicates that under high temperatures and high electric fields,

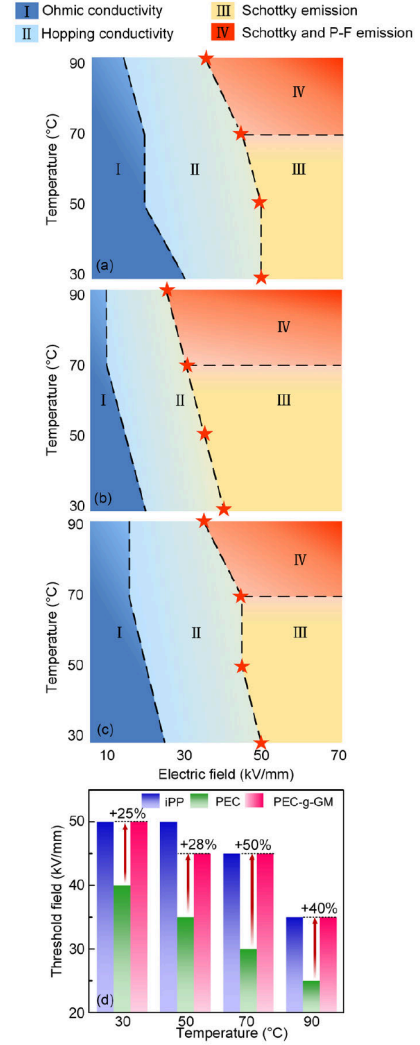


Fig. 8. Schematic of the dominant charge conductivity mechanism that varies with temperature and electric field. (a) iPP. (b) PEC. (c) PEC-g-GM. (d) Threshold electric fields of samples.

compared to PEC, it is more difficult for charges to be injected into PEC-g-GM, and charges trapped in traps are also less likely to escape and form a current. The threshold electric fields, at which the dominant conductivity mechanism in the samples shifts from hopping conductivity to Schottky and P-F emission, are compared in Fig. 8(d). It can be seen that the threshold electric fields of PEC-g-GM are almost the same as those of iPP and are much higher than those of PEC at all temperatures. This indicates that grafting GM will make it more difficult for charges to penetrate the Schottky barrier and to be injected into the interior of the sample, causing charges to accumulate at the interface between the electrode and the sample, thereby reducing conductivity.

#### E. DCIC-Q(t) Characteristics

Based on the DCIC- $Q(t)$ , the dynamic charge characteristics of the samples under electric fields from 1 to 70 kV/mm were obtained. In order to highlight the dynamic charge changes and the charge injection process of the samples, the final charge  $Q(t_m)$  is divided by  $Q_0$  to obtain the  $Q(t_m)/Q_0$

curves and the discharge process is omitted. The variation of  $Q(t_m)/Q_0$  with time is shown in Fig. 9.

The initial charge  $Q_0$  of different samples shows little variation. This is because the initial charge results from the displacement current charging the capacitance between the plate and plate electrode structure. Since  $Q_0$  is highly related to the geometry of the electrodes, it remains nearly the same across different samples. The accumulation of conductive charge is mainly related to the slope of the  $Q(t)$  curve after it stabilizes. As shown in Fig. 9, at different temperatures, the slope for PEC-g-GM is always smaller than that of PEC, indicating a slower accumulation rate of conductive charge in PEC-g-GM. This is also consistent with Fig. 4.

It can be seen that  $Q(t_m)/Q_0$  at the end of each 10-min polarization tends to increase with the increase of applied voltage. Moreover, at any temperature and electric field,  $Q(t_m)/Q_0$  of PEC-g-GM is smaller than that of PEC. At a low temperature of 30 °C and an electric field of 70 kV/mm, when polarization ends,  $Q(t_m)/Q_0$  of PEC-g-GM is 5.97% lower than that of PEC. At a high temperature of 90 °C and an electric field of 70 kV/mm, when polarization ends,  $Q(t_m)/Q_0$  of PEC-g-GM is 31.07% lower than that of PEC. Under a low electric field,  $Q(t_m)/Q_0$  of PEC-g-GM is close to that of PEC, but it is much smaller than PEC under a high electric field, indicating that the influence of electric field on PEC-g-GM is very small. This indicates that the grafted GM effectively suppresses charge injection and accumulation under high temperature and high electric field.

In order to focus on the space charge properties of the samples, the absorbed charge  $Q_{\text{abs}}$  of the samples after polarization at different electric fields and temperatures was obtained by fitting through (3). The results are shown in Fig. 10. From Figs. 4 and 7, it can be seen that there is significant space charge injection only when the electric field is greater than 30 kV/mm, so the fit starts from 30 kV/mm. At a low temperature of 30 °C and an electric field strength of 70 kV/mm,  $Q_{\text{abs}}$  of PEC is 78.95 nC, while that of PEC-g-GM is 55.56 nC, representing a 29.6% reduction compared to PEC. At a high temperature of 90 °C and an electric field strength of 70 kV/mm,  $Q_{\text{abs}}$  of PEC is 194.66 nC, while that of PEC-g-GM is 130.03 nC, representing a 33.2% reduction compared to PEC. As the temperature and electric field increase, the difference in  $Q_{\text{abs}}$  between PEC-g-GM and PEC becomes more pronounced, indicating that the role of GM is significant under high temperature and high electric field.

### F. Trap-Level Distribution

The trap-level distribution for each sample is shown in Fig. 11. The temperature and current coordinates correspond to the trap energy level and trap density, respectively. The  $\beta$  peak appears in the temperature range of −25 °C to 10 °C and corresponds to the orientation polarization of the material. As the temperature rises, PP undergoes a transition from the glassy state to the rubbery state. This transition enhances the mobility of molecular chains in the amorphous region, making them more susceptible to dipole polarization, which results in the formation of the  $\beta$  current peak [8]. Since the glass transition temperatures of iPP and POE used in this experiment are close to each other, there is only one peak for PEC at low

temperatures. In addition, the  $\beta$  peak values of PEC and PEC-g-GM are both higher than that of PP, partly because POE mostly exists in the amorphous phase, with looser molecular chains, and is more prone to polarization under an electric field. On the other hand, it is because GM is a polar molecule with a larger dipole moment.

With further temperature rise, charges trapped in shallow traps within the amorphous region are first released, resulting in the  $\alpha_1$  shallow trap current peak. It can be seen that the  $\alpha_1$  peak of PEC is much higher than that of iPP and PEC-g-GM. This is due to the fact that the addition of POE increases the proportion of amorphous phase in the overall structure of iPP. The amorphous phase has the characteristic of irregular arrangement of molecular chains, which will produce more shallow traps. In contrast, the  $\alpha_1$  peak of PEC-g-GM was close to that of iPP, suggesting that the grafted GM reduced the number of shallow traps in PEC. This is mainly because grafting GM leads to smaller spherulites in PEC, resulting in more crystalline interfaces. The shallow traps originally located in the amorphous regions are transformed into deep traps at the crystalline interfaces, thereby reducing the number of shallow traps.

As the temperature continues to increase, charges trapped in deep traps within the crystalline region are also released, giving rise to the  $\alpha_2$  deep trap current peak. The  $\alpha_2$  peak of PEC is located at 0.90 eV, while the peaks  $\alpha_2$  of iPP and PEC-g-GM are located at 0.95 and 0.97 eV, respectively, both exhibiting higher trap densities. This indicates that the grafted GM introduced numerous deep traps inside the sample. These deep traps come from the polar groups of GM and the interface between spherulites. These deep traps can effectively capture charges at the electrode-sample interface, forming a homocharge blocking layer that suppresses charge injection, thereby increasing the threshold electric field  $E_T$ .

## IV. DISCUSSION

The band structure of the molecules was simulated through quantum chemistry calculations. Modeling calculations were performed in Materials Studio for PP, POE, and PP grafted GM insulation (PP-g-GM). The degree of polymerization of ethylene in PP and PP-g-GM is 10. The POE carbon backbone has 20 carbon atoms and two octene side chains are present. The hybridized generalized B3LYP was used to optimize the molecule and calculate the energy-level structure. During the geometry optimization process, the molecule was iteratively calculated using the conjugate gradient method to obtain the lowest energy conformation. The energy convergence criterion was set to  $1 \times 10^{-5}$  Ha, the force convergence criterion was set to  $2 \times 10^{-3}$  Ha, the displacement convergence criterion was set to  $5 \times 10^{-3}$  Ha, and the maximum number of optimization cycles was 100. From Fig. 12, it is clear that the lowest unoccupied molecular orbital (LUMO) energy level of PP is 1.873 eV and the highest occupied molecular orbital (HOMO) energy is −7.808 eV. Then, the corresponding electron affinity  $\chi(\text{PP})$  of PP is −1.873 eV, the ionization energy  $\Phi_i(\text{PP})$  is 7.808 eV, and the bandwidth  $\Phi_g(\text{PP})$  is 9.681 eV. In contrast, the bandwidth  $\Phi_g$  of POE is 9.132 eV and the ionization energy  $\Phi_i$  is 7.511 eV, both of which are smaller than PP.

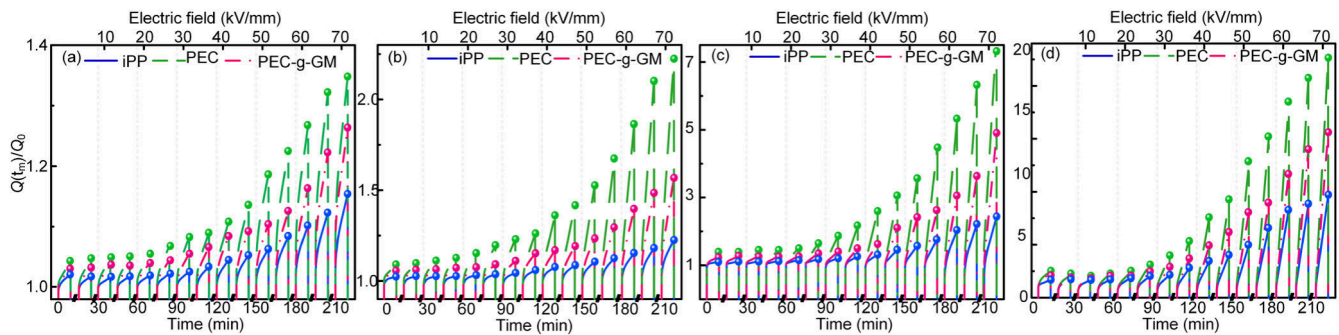


Fig. 9.  $Q(t_m)/Q_0$  curves of samples. (a) 30 °C. (b) 50 °C. (c) 70 °C. (d) 90 °C.

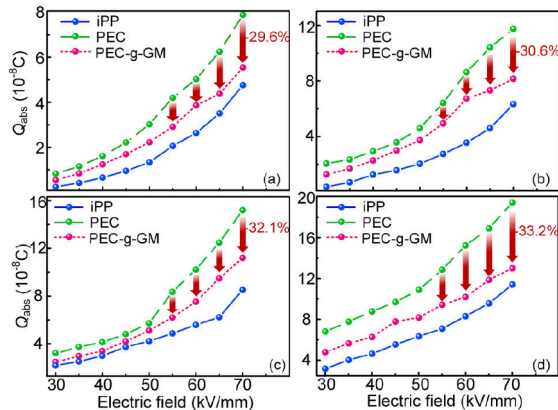


Fig. 10.  $Q_{abs}$  of samples at different electric fields and temperatures. (a) 30 °C. (b) 50 °C. (c) 70 °C. (d) 90 °C.

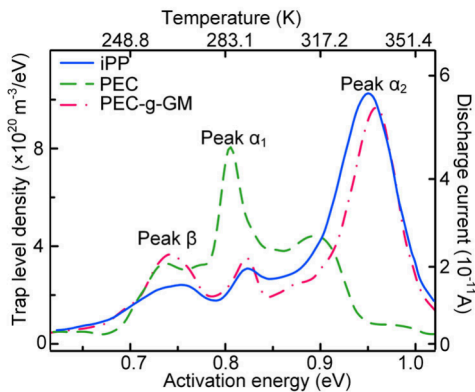


Fig. 11. TSDC measurement current curves of the three samples.

From the perspective of energy band structure, POE is more prone to electronic transitions than PP; therefore, the insulation performance of PEC is worse than PP. Grafted GM introduced localized state traps into PP, with electron traps reaching a maximum depth of 2.274 eV and hole traps reaching a maximum depth of 1.873 eV. These localized state traps inhibit carrier migration within the composite insulating material.

This study also constructs amorphous region unit cells for the three systems. The mass ratio of PP to POE in PEC is 7:3, and a grafted molecular chain is introduced in PEC-g-GM, as shown in Fig. 13. The intermolecular interaction potentials of three structures were calculated, as shown in Table II.

It can be found that the van der Waals (VdW) potential of PEC decreases compared to PP because the random chain segments

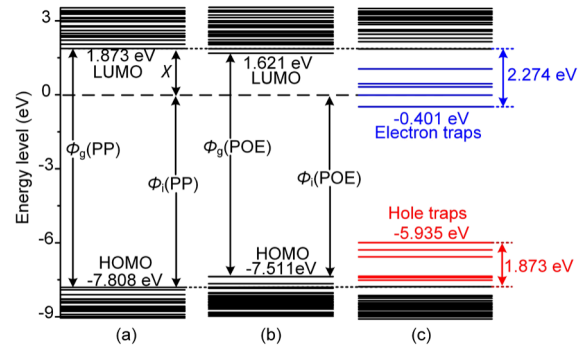


Fig. 12. Energy levels of (a) PP, (b) POE, and (c) PP-g-GM.

TABLE II  
SIMULATION RESULTS OF INTERMOLECULAR  
INTERACTION POTENTIAL ENERGY

| Sample   | VdW potential (kJ/mol) | ES potential (kJ/mol) |
|----------|------------------------|-----------------------|
| PP       | -2937.09               | -7.72                 |
| PEC      | -2660.82               | -5.36                 |
| PEC-g-GM | -2726.21               | -7.08                 |

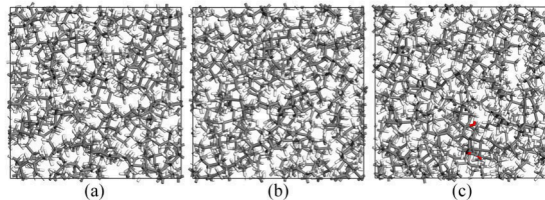


Fig. 13. Amorphous unit cell. (a) PP. (b) PEC. (c) PEC-g-GM.

reduces the VdW effect. In addition, the addition of POE reduces the degree of dense stacking between PP, resulting in a decrease in the electrostatic potential (ES potential). The VdW potential energy and ES potential energy of PEC increased after the graft modification, which was attributed to the formation of dipole interactions by the introduced polar chain segments, which enhanced the electrostatic interactions of the system. Thus, grafting GM molecules increases the intermolecular interactions and reduces the total energy of the amorphous system, making the system more stable.

## V. CONCLUSION

- 1) The leakage current of graft-modified insulating PEC-g-GM is much smaller than that of PEC in the range of 30 °C–90 °C. At a temperature of 90 °C and a field of 70 kV/mm, the leakage current of PEC-g-GM is 33.4% lower than that of PEC. Moreover, the Schottky emission threshold field of PEC-g-GM is higher than that of PEC.
- 2) The space charge of PEC-g-GM decreased by 31.07% compared to PEC at 90 °C and 70 kV/mm. Meanwhile, grafting GM increases the trap energy level and trap density of PEC, which effectively suppresses the carrier migration.
- 3) Grafting GM introduces localized state traps in the PP energy levels and enhances the intermolecular interaction potential. After the charge jumps over the potential barrier, it gets trapped at the interface between the material and the electrode, thereby suppressing the injection of space charge.

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